

Bachelor of Technology — Mathematics and Computing

Table 1: Mathematics and Computing — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	51
	Discipline Electives	15
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Mathematics and Computing — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
2	CS-214	Computer Organisation	3-0-2-4	4
3	MA-210	Real and Complex Analysis	2.5-0.5-0-3	3
4	MA-211	Ordinary Differential Equations	3-1-0-4	4
5	CS-304	Formal Language and Automata Theory	3-0-0-3	3
6	MA-220	Partial Differential Equations	3-1-0-4	4
7	MA-221	Numerical Analysis	3-1-0-4	4
8	MA-222	Applied Linear Programming	3-1-0-4	4
9	CS-212	Design of Algorithms	3-0-2-4	4
10	MA-310	Matrix Computation and Lab	3-0-2-4	4
11	MA-311	Mathematical Modelling	3-0-0-3	3
12	MA-322	Applied Graph Theory		4
13	MA-321	Numerics of Differential Equation		4
14	MA-323P	Applied Databases Practicum	0-0-3-2	2
Total				51

Table 3: B. Tech Mathematics and Computing — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Mathematics and Computing — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Mathematics and Computing — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
4	DC	CS-214	Computer Organisation	3-0-2-4	4
5	DC	MA-210	Real and Complex Analysis	2.5-0.5-0-3	3
6	DC	MA-211	Ordinary Differential Equations	3-1-0-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Mathematics and Computing — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-304	Formal Language and Automata Theory	3-0-0-3	3
2	DC	MA-220	Partial Differential Equations	3-1-0-4	4
3	DC	MA-221	Numerical Analysis	3-1-0-4	4
4	DC	MA-222	Applied Linear Programming	3-1-0-4	4
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Mathematics and Computing — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-212	Design of Algorithms	3-0-2-4	4
2	DC	MA-310	Matrix Computation and Lab	3-0-2-4	4
3	DC	MA-311	Mathematical Modelling	3-0-0-3	3
4	DC	MA-322	Applied Graph Theory		4
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Mathematics and Computing — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	MA-321	Numerics of Differential Equation	3-0-2-4	4
2	DC	MA-323P	Applied Databases Practicum	0-0-3-2	2
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
4	DC	MA-322	Applied Graph Theory	3-0-2-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Mathematics and Computing — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	MC-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Mathematics and Computing — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	MC-010	Internship		2
2	MTP	MC-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 15 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

This programme groups its Discipline Electives into the named baskets below; the remaining approved electives follow as a general list.

Table 11: Mathematics and Computing — Discipline Elective Basket I — Foundation Module (choose from these)

Sl. No	Course Code	Course Name	Credits
1	MA-251	Abstract Algebra	3
2	MA-252	Functional Analysis	3
3	MA-253	Measure Theory	3
4	MA-254	Topology	3
5	MA-255	Number Theory	3

Table 12: Mathematics and Computing — Discipline Elective Basket II — Advanced Modelling Module (choose from these)

Sl. No	Course Code	Course Name	Credits
1	MA-351	Climate Modelling	3
2	MA-352	Computational Financial Modelling & Lab	3
3	MA-353	Modelling of Infectious Disease	3
4	MA-354	Mathematical Image Processing	3
5	MA-355	Mathematical Control Theory	3
6	MA-356	Modelling and Simulation	3
7	MA-357	Modelling Population Dynamics	3
8	MA-546	Introduction to Quantitative Finance	4
9	MA-570	Data Driven Dynamical Systems	3

Table 13: Mathematics and Computing — Other Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BE-303	Applied Biostatistics	4
2	BE-304	Bioinformatics	4
3	CE-251	Hydraulics Engineering	3
4	CS-241	Introduction to Cryptography	3
5	CS-302	Paradigms of Programming	4
6	CS-306	Operating Systems	3
7	CS-309	Information and Database Systems	4
8	CS-313	Computer Networks	4
9	CS-315	Artificial Intelligence (CS-362, CS-562)	4
10	CS-502	Compiler Design	4
11	CS-514	Advance Data Structure and Algorithm (CS-580)	4
12	CS-563	Scalable Data Science	4
13	CS-606	Cognitive Modeling	3
14	CS-609	Speech processing	3
15	CS-677	Soft Computing	3
16	DS-201	Data handling and visualization	3
17	DS-312	Computing Systems for Data Processing	4
18	DS-416	Fundamentals of Machine Learning (CS-669)	4
19	EE-203	Network Theory	3
20	EE-260	Signals and Systems	3
21	EE-301	Control Systems	3
22	EE-511	Computer Vision	4
23	EE-530	Optimization theory	3
24	EE-608	Digital Image Processing	4
25	MA-001	Preparatory Mathematics - 1	4
26	MA-002	Preparatory Mathematics - 2	4

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
27	MA-101	Calculus I	4
28	MA-102	Linear Algebra	4
29	MA-120	Introduction to Computing Systems and Databases	4
30	MA-201	Probability and Statistics	4
31	MA-460	Nonlinear Dynamics and Chaos	3
32	MA-465	Ordinary Differential Equations	3
33	MA-510	Climate Change Analysis	3
34	MA-516	Topology	4
35	MA-521	Topology and Functional Analysis	4
36	MA-522	Partial Differential Equations	4
37	MA-523	Numerical Analysis	4
38	MA-524	Probability and Statistics	4
39	MA-525	Heuristic Optimization	3
40	MA-526	An Introduction to Wavelets	3
41	MA-527	Field and Galois Theory	3
42	MA-528	Measure Theory and Integration	4
43	MA-529	Statistical Inference	4
44	MA-530	Graph Theory	4
45	MA-546	Introduction to Quantitative Finance	4
46	MA-548	Abstract Algebra	4
47	MA-549	Abstract Algebra	4
48	MA-550	Statistical Data Analysis	3
49	MA-551	Numerical Analysis	3
50	MA-552	Number Theory	3
51	MA-553	Mathematical Foundations of Financial Engineering	3
52	MA-555	Introduction to Partial Differential Equations for Engineers	3
53	MA-560	Nonlinear Dynamics and Chaos	3
54	MA-563	Statistical Computation and Simulation	4
55	MA-565	Numerical Methods in Quantitative Finance	3
56	MA-568	Real Analysis	3
57	MA-570	Data-driven Dynamical Systems	3
58	MA-571	Introduction to Approximation algorithms	4
59	MA-572	Applied Multivariate Statistical Analysis	4
60	MA-581P	Post Graduate Project I	6
61	MA-582P	Post Graduate Project-2	8
62	MA-588	Mathematical Control Theory	4
63	MA-595	Calculus as Ganita	3
64	MA-601	Real and Functional Analysis	3
65	MA-603	Advanced Partial Differential Equations	3
66	MA-604	Introduction To Game Theory	3

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
67	MA-605	Statistical Data Analysis	3
68	MA-607	Numerical Analysis	3
69	MA-608	Computational Fluid Dynamics	3
70	MA-609	Numerics of Partial Differential Equations	3
71	MA-610	Mathematical Modeling	3
72	MA-611	Statistical tools and Computing	4
73	MA-612	Operator Theory	4
74	MA-621	Modeling Population Dynamics	3
75	MA-641	Operations Research	3
76	MA-644	Dynamical Systems	3
77	MA-650	Mathematical Models for Infectious Diseases	3
78	MA-651	Optimization Techniques	3
79	MA-652	Stability Theory of Differential Equations	3
80	MA-653	Computational Financial Modelling	3
81	MA-653P	Computational Financial Modelling Lab	1
82	MA-654	Financial Engineering	3
83	MA-655	Fixed Income Securities	3
84	MA-656	Stochastic Calculus for Financial Engineering	3
85	MA-665	Semigroup Of Bounded Linear Operators	3
86	MA-704	Dynamical System	3
87	MA-709	Numerical Linear Algebra	3
88	MA-765	Fractional Differential Equations	4
89	MA-780	Topics on Semigroup Theory	3
90	ME-210	Fluid Mechanics	3
91	ME-213	Engineering Thermodynamics	4
92	ME-303	Heat Transfer	3
93	ME-308	Manufacturing Engineering	3
94		Time Series Analysis	3
95		Mathematical Method for Signal Processing	4
96	CS-669	Pattern Recognition	4
97	BE-301	Biomechanics	4
98	BY-507	Genetic Engineering	4
99	CS-352	Transportation Engineering	3
100	CS-352P	Transporting Engg. Lab	1
101		Information Systems and Databases	4
102		Times Series Analysis and Applications / Bayesian Data Analysis and Applications	3
103		Big Data: Management and Analytics	4
104	EP-302	Computational Methods for Engineering	3
105	ME-310	System Dynamics and Control	3
106	ME-210P	Fluid Mechanics Lab	1

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
107	ME-303P	Heat Transfer Lab	1
108		Cellular Automata	3
109	CS-606	Computational Modeling of Social Systems	3
110	MA-562	An Introduction to Bayesian Analysis	4
111	MA-561	Fluidics and Microfluidics	4
112	MA-573	Advanced Mathematical Foundations of Machine Learning	4
113	MA-574	An Introduction to Probabilistic Machine Learning	4
114	MA-576	Nonlinear Optimization	4
<i>Compulsory project components — counted as DE, not selectable</i>			
115	MC-498P	Major Technical Project - I (compulsory; counted as DE)	4
116	MC-499P	Major Technical Project - II (compulsory; counted as DE)	4